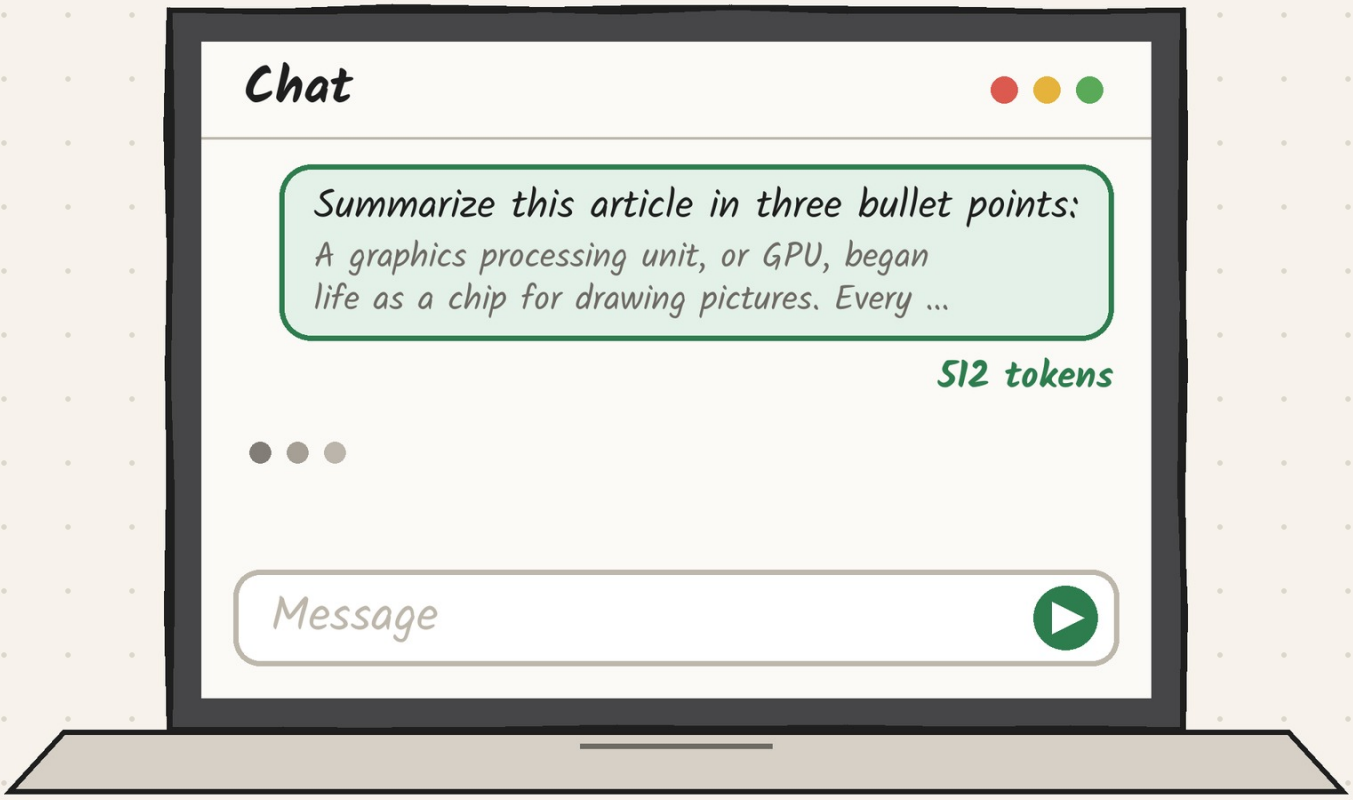
AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100



the prompt travels from your laptop to the H100 server

(network time is not counted: every time in this video is measured on the server)

1/13

It starts with your prompt

an article to summarize: 512 tokens

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how Qwen3.8-27B makes tokens on an H100



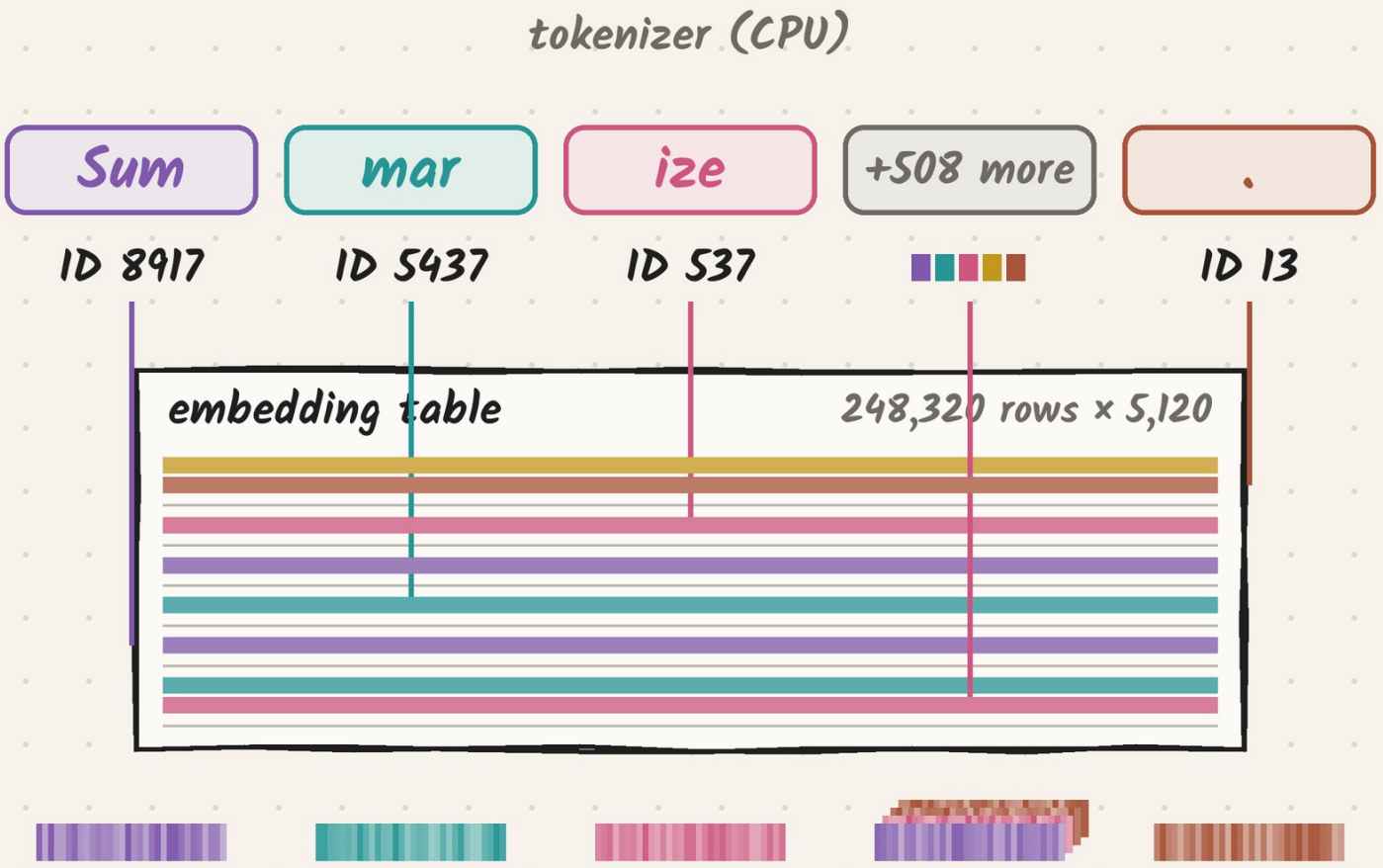
2/13

The tokenizer (CPU) splits it

into 512 token IDs: Sum · mar · ize …

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how Qwen3.8-27B makes tokens on an H100



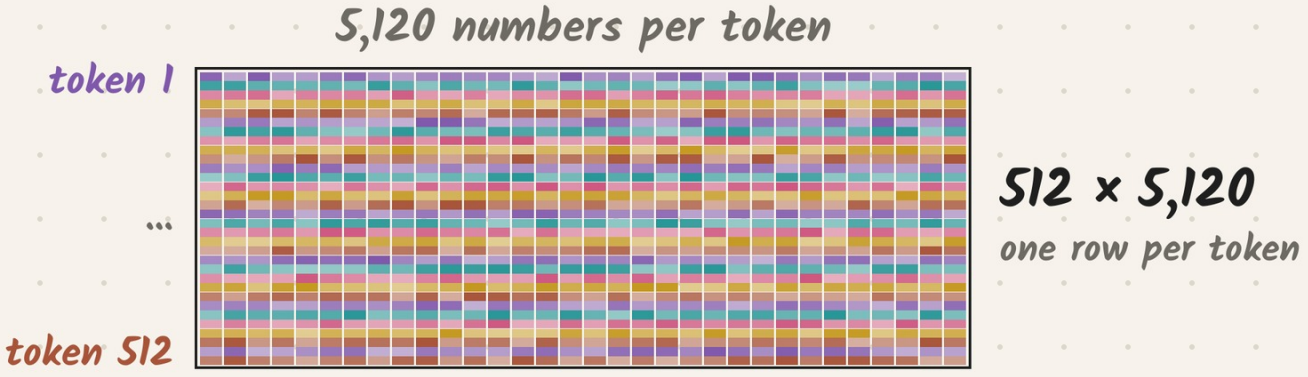
3/13

Each ID picks a row of 5,120 numbers

from the embedding table

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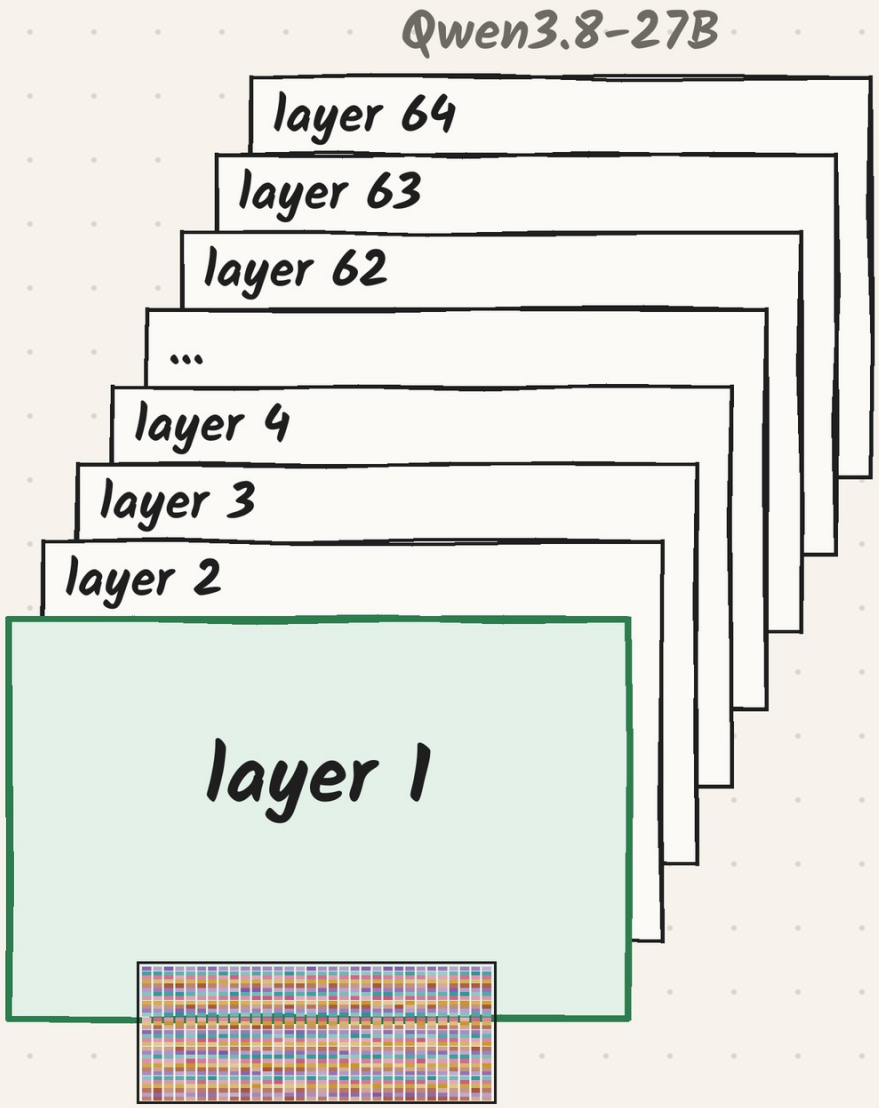
4/13

Stack them: 512 × 5,120

one row per token

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how Qwen3.8-27B makes tokens on an H100



513

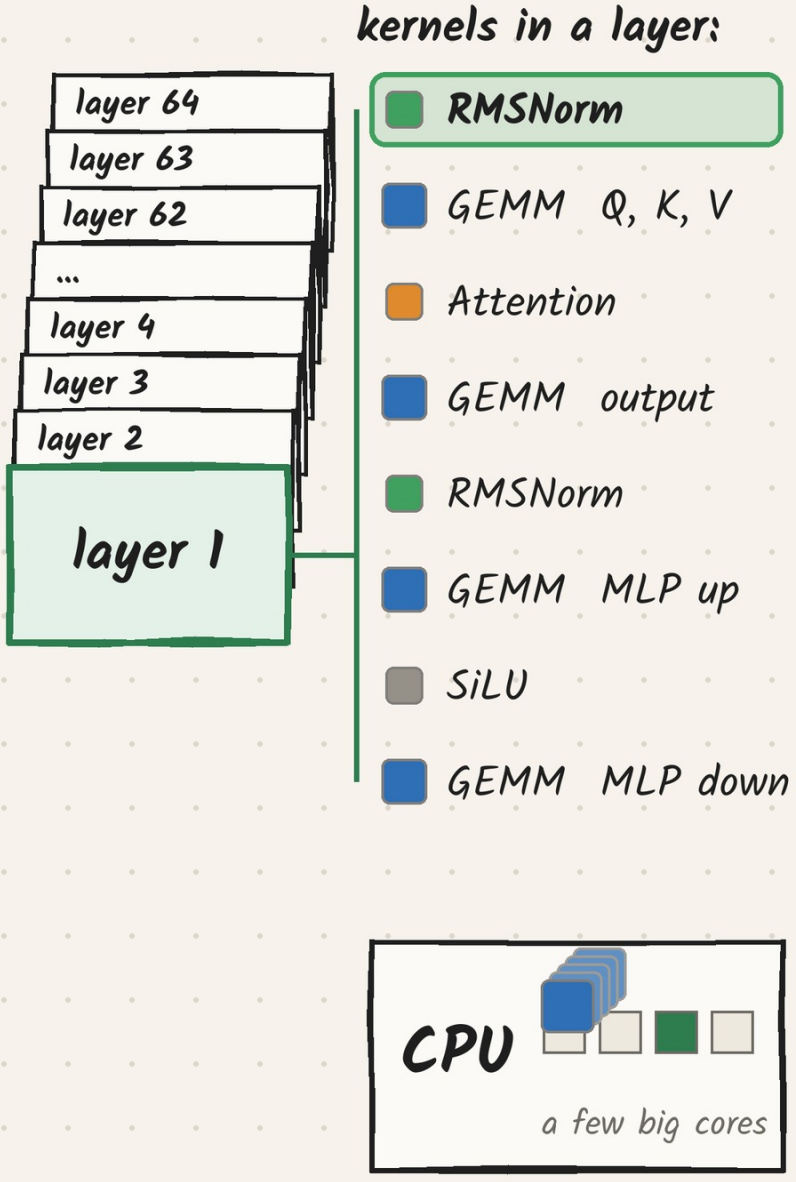
Prompt phase: all 512 go in together

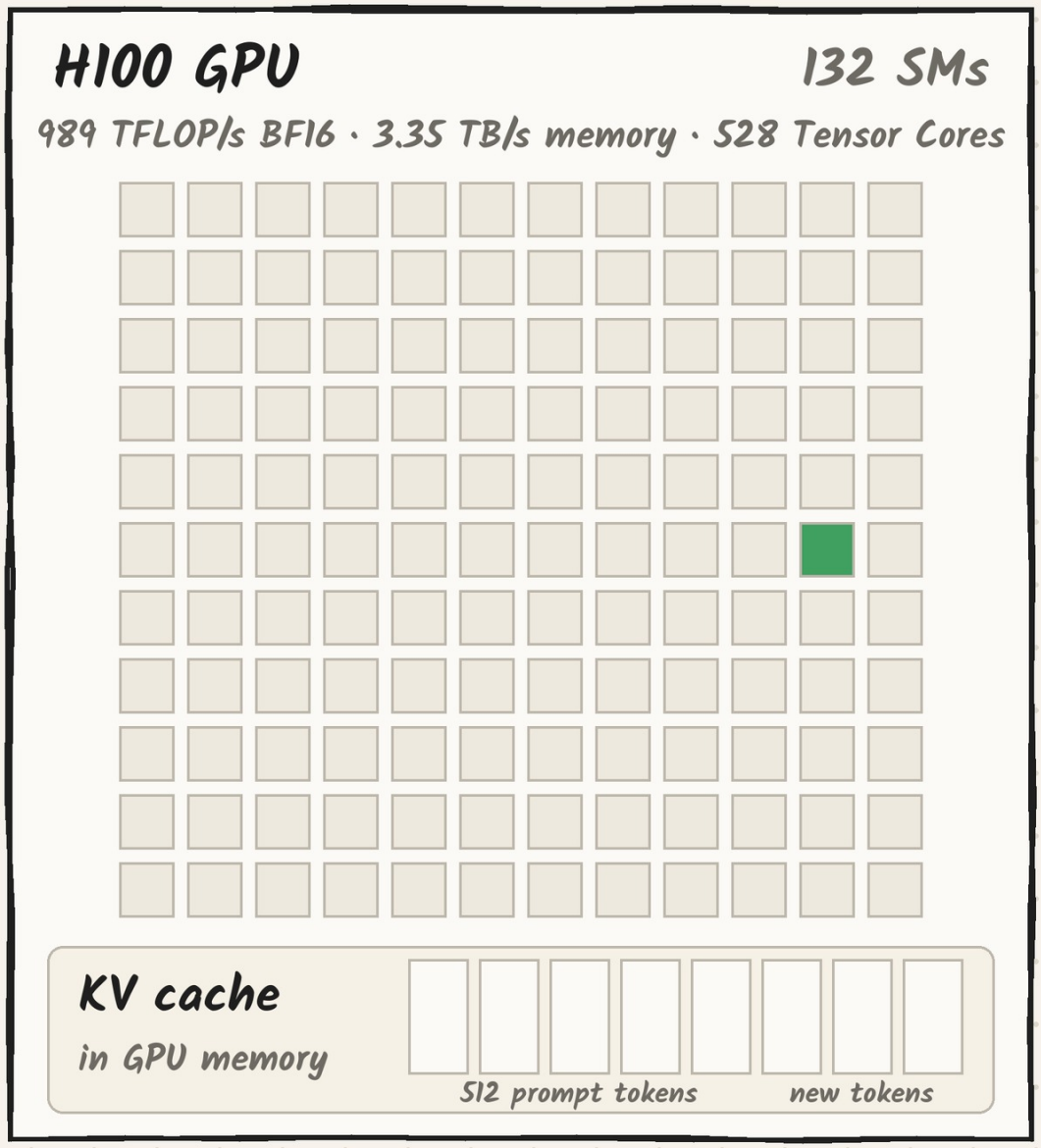
the whole matrix enters layer 1

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AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100





6/13

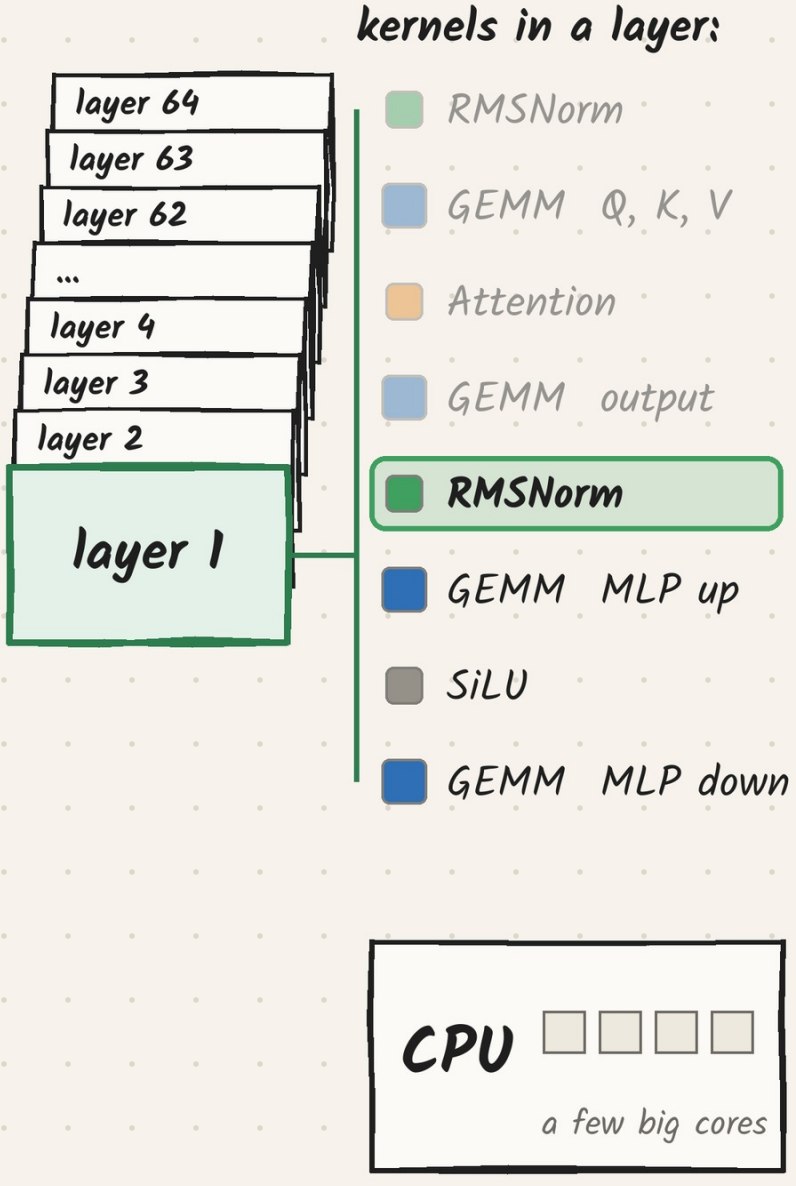
Layer 1 = a chain of kernels

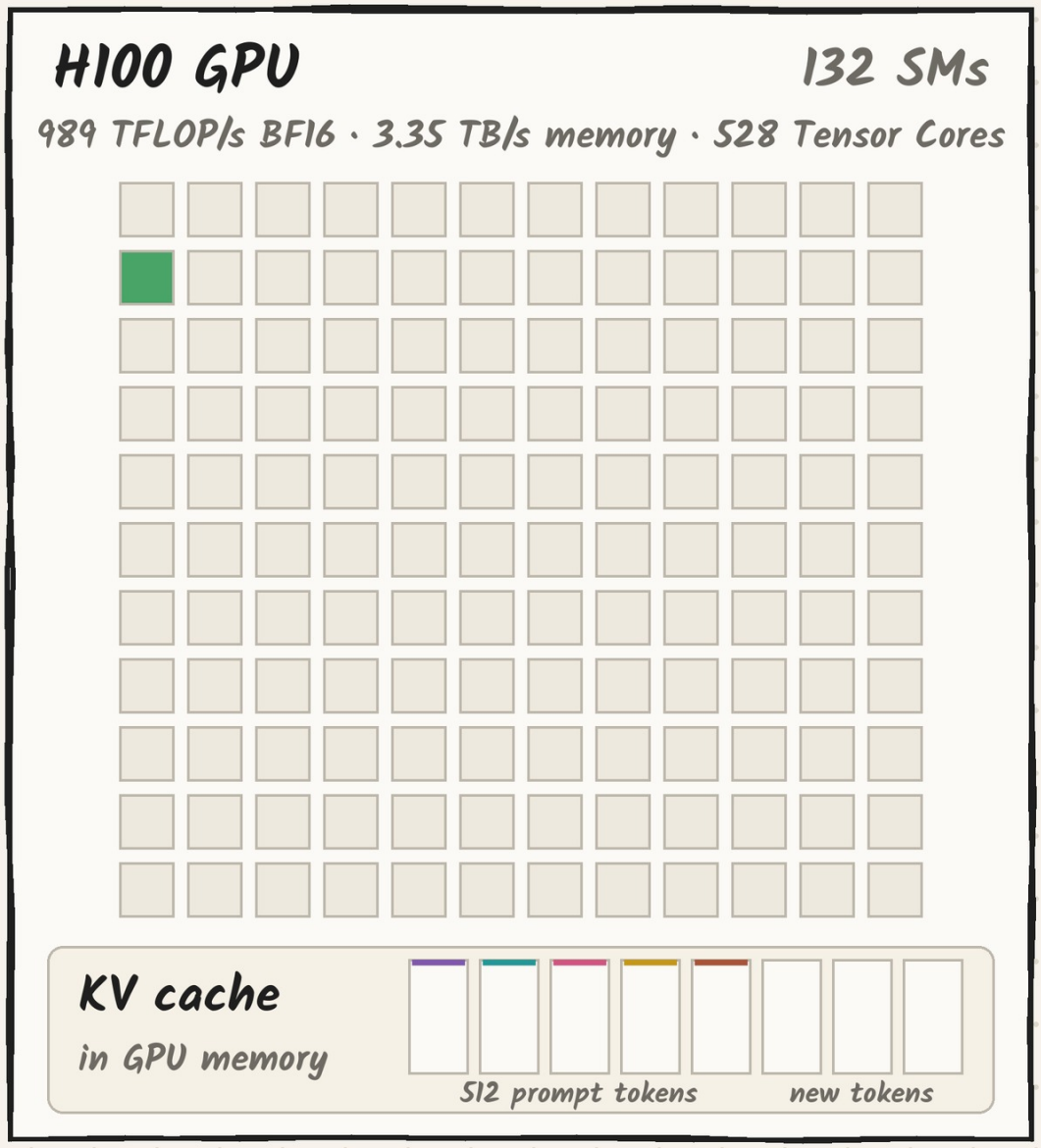
the CPU sends them, the GPU runs them in order

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AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100





7/13

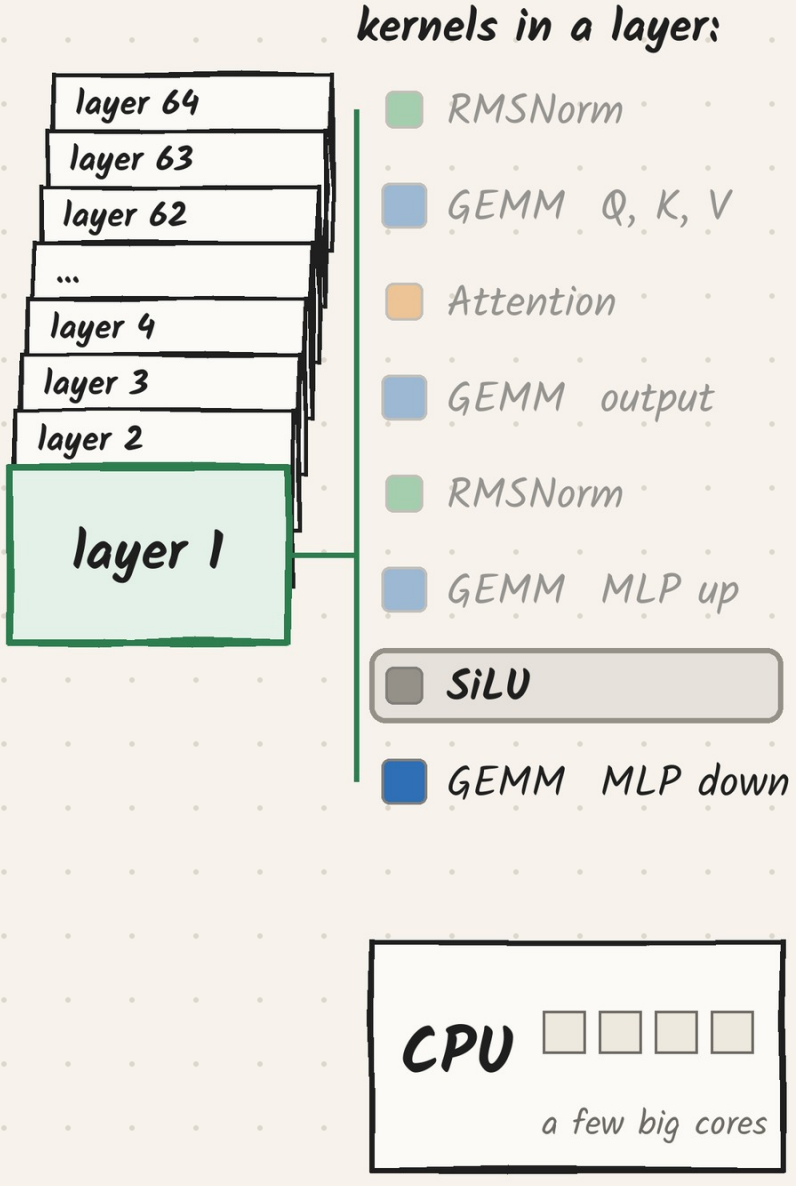
Q, K, V GEMM saves K and V

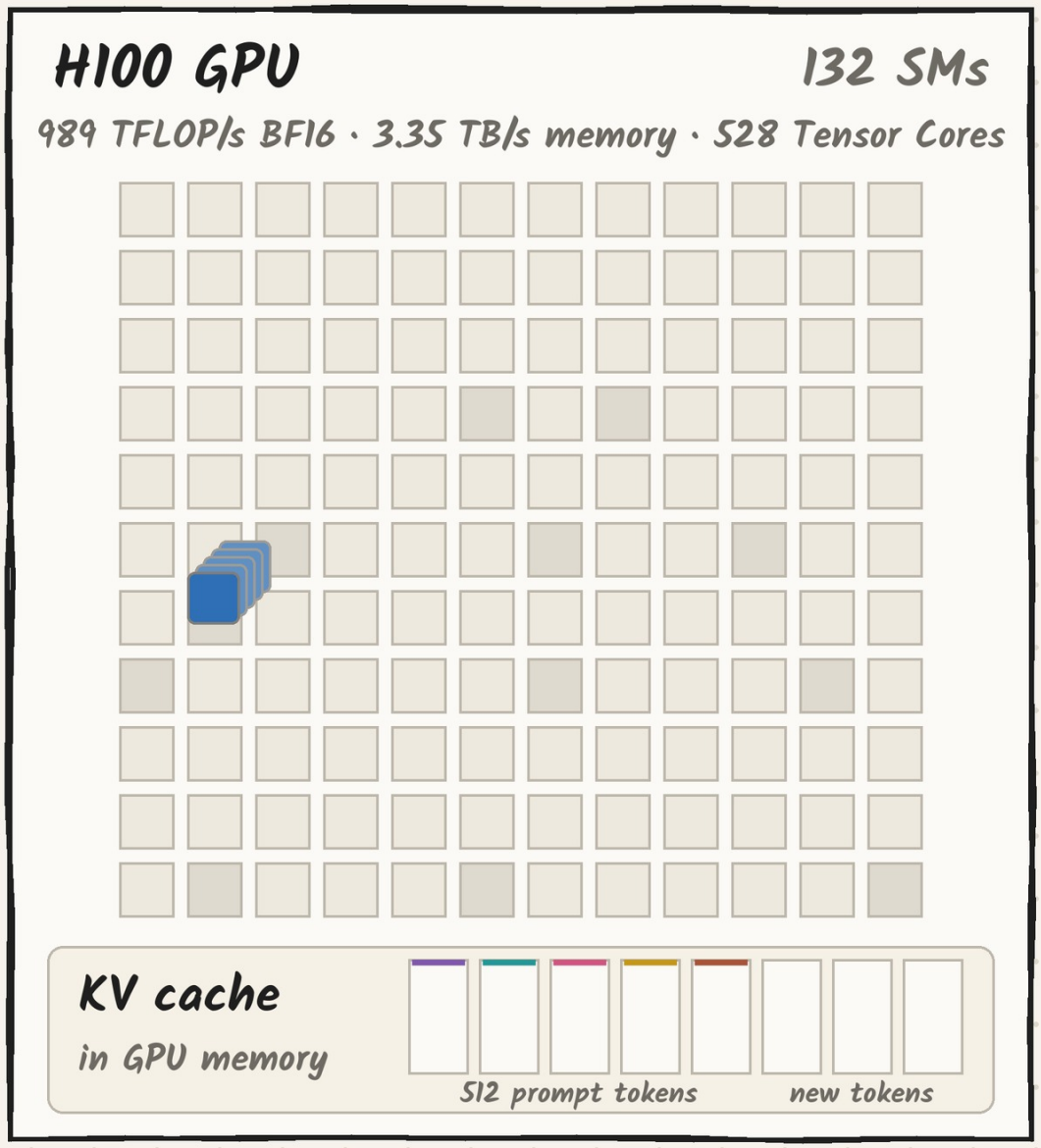
of all 512 tokens, in the KV cache

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AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100





8/13

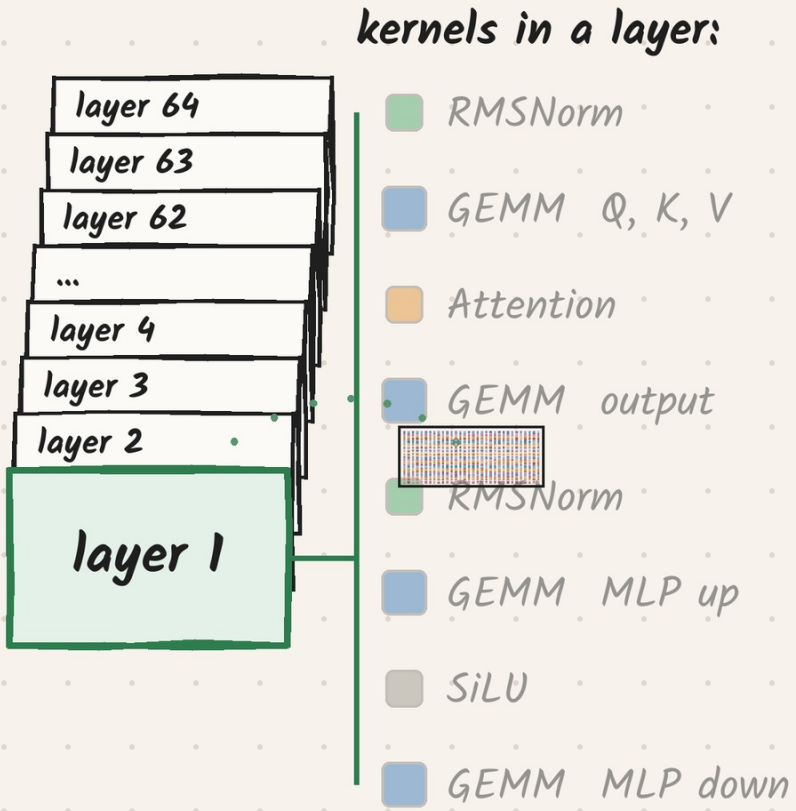
Each kernel spreads over 132 SMs

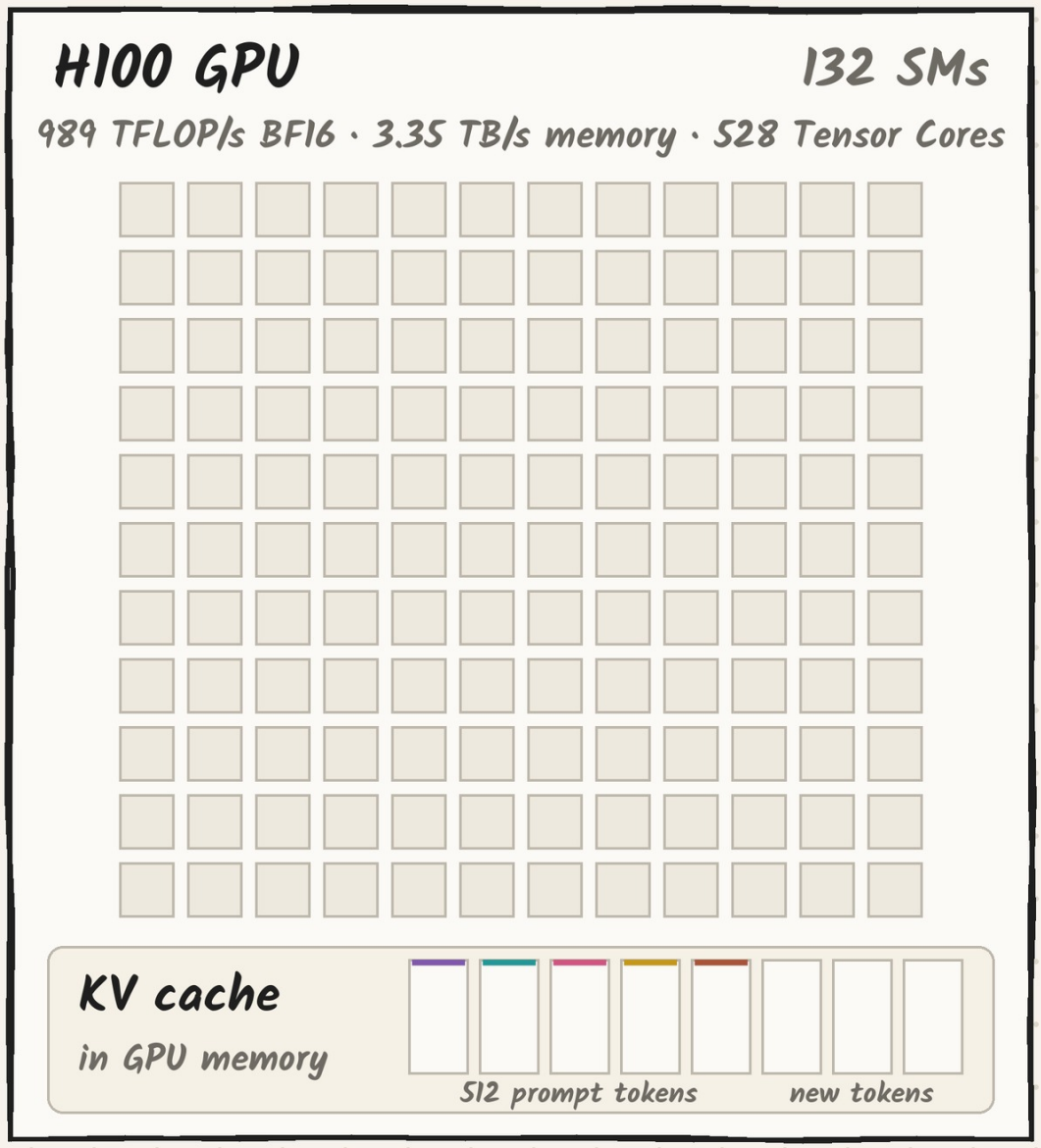
GEMMs (blue): 92.6% of the GPU time

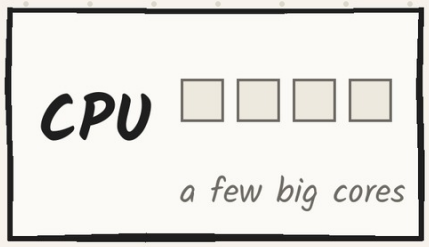
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AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100







9/13

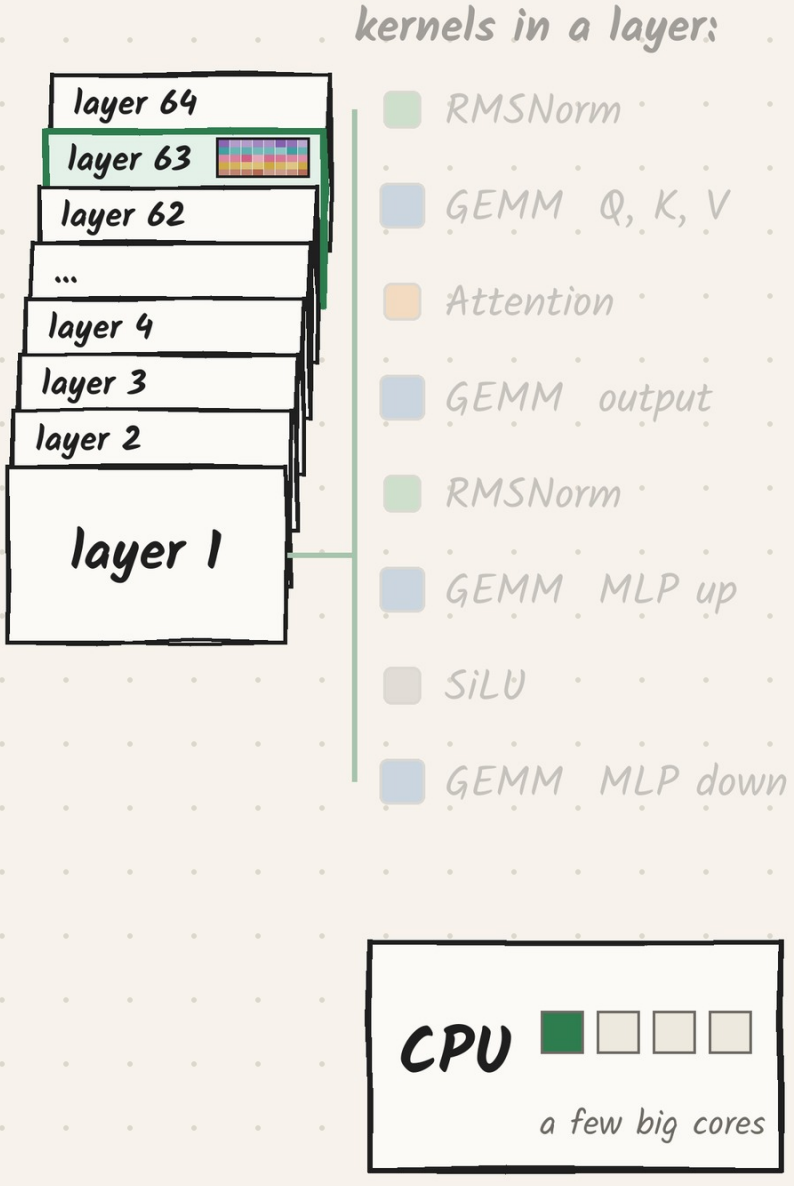
Layer 1's output = layer 2's input

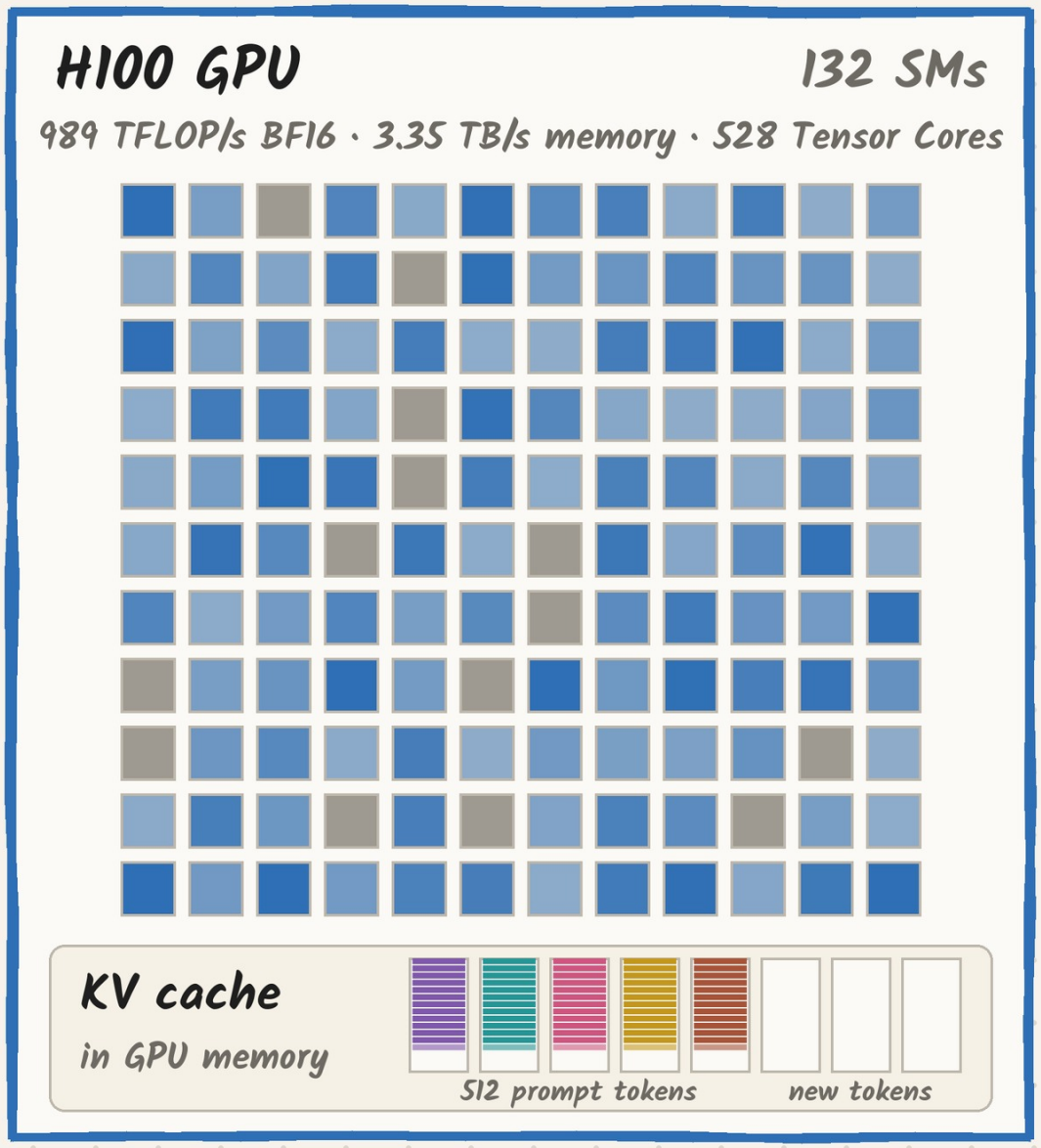
still 512 × 5,120

Qwen3.8-27B · vLLM · 1x H100 SXM · specs: NVIDIA H100 datasheet, Qwen config & tokenizer

AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100





10/13

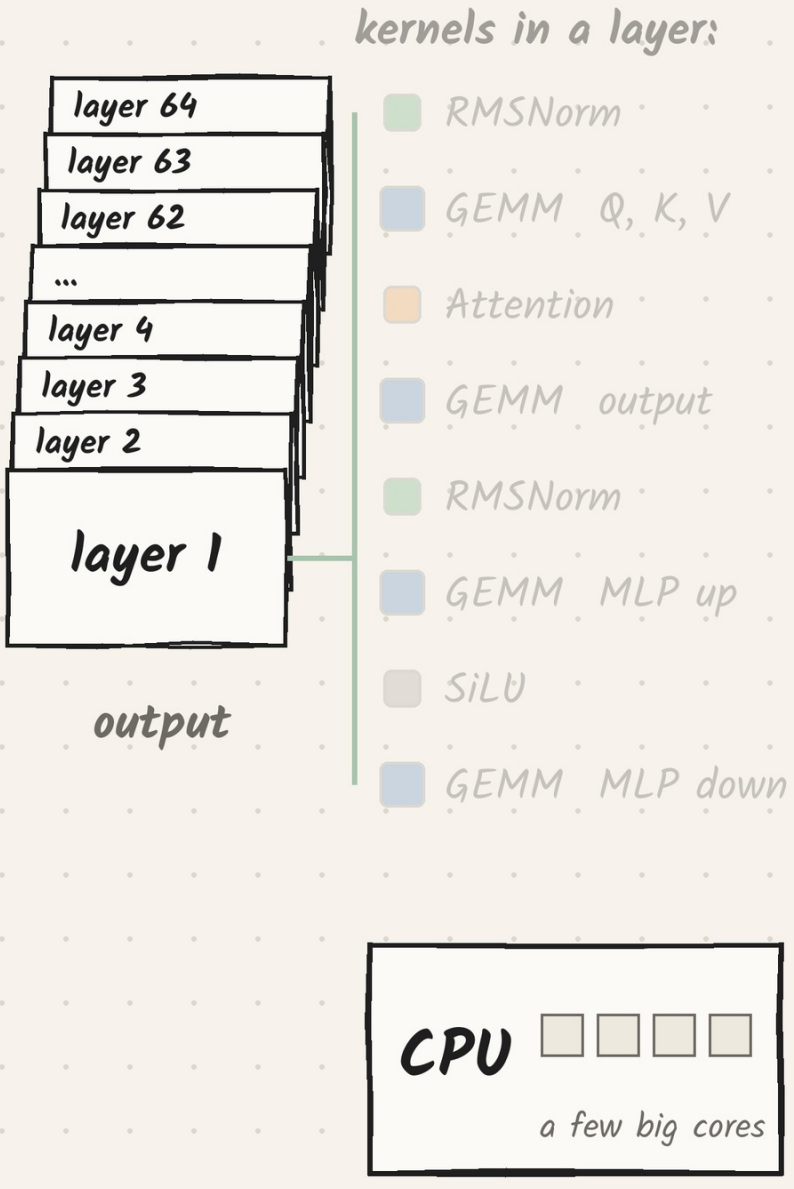
Then layers 2, 3 … 64

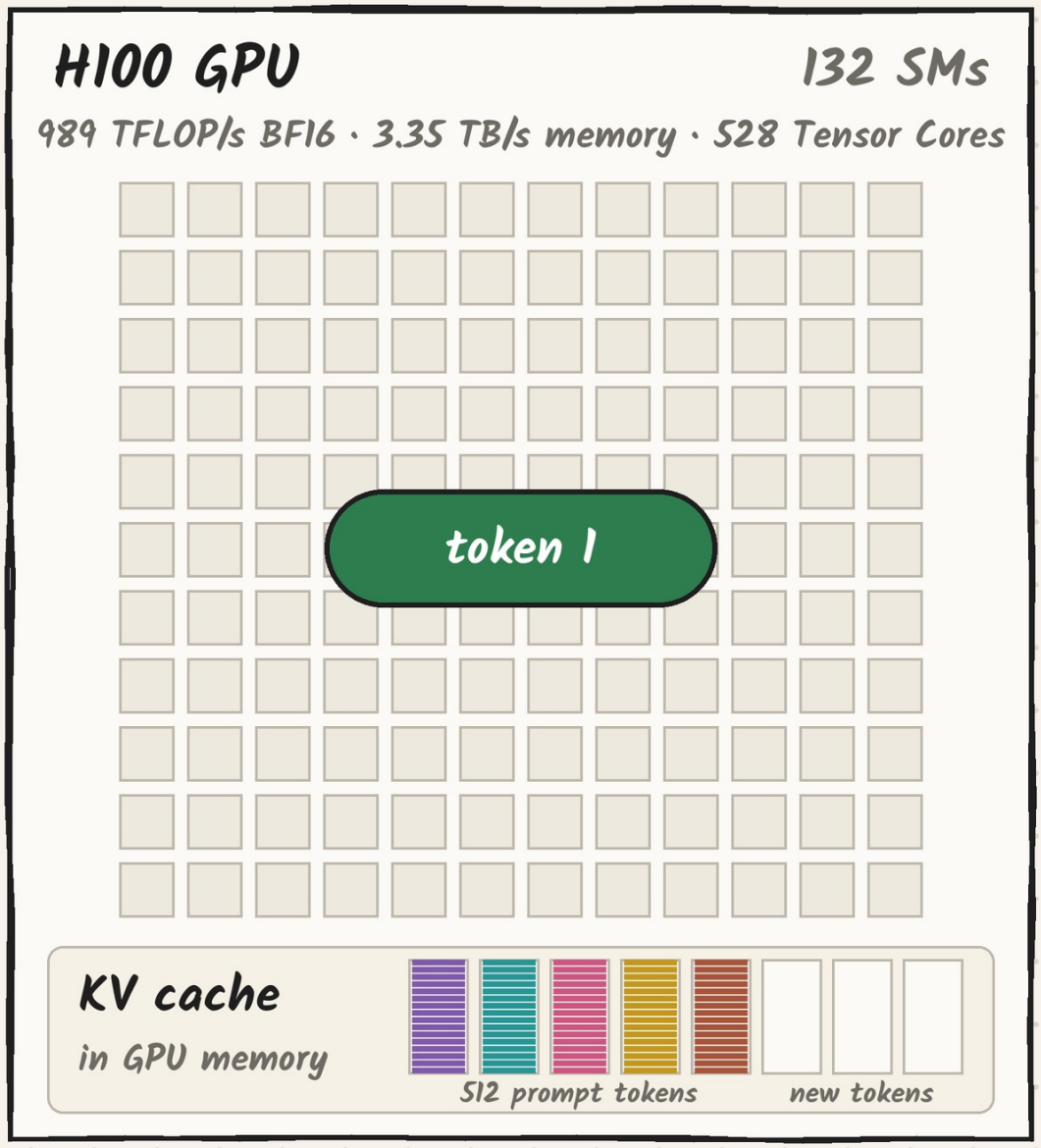
now: layer 63

Qwen3.8-27B · vLLM · 1x H100 SXM · specs: NVIDIA H100 datasheet, Qwen config & tokenizer

AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100





11/13

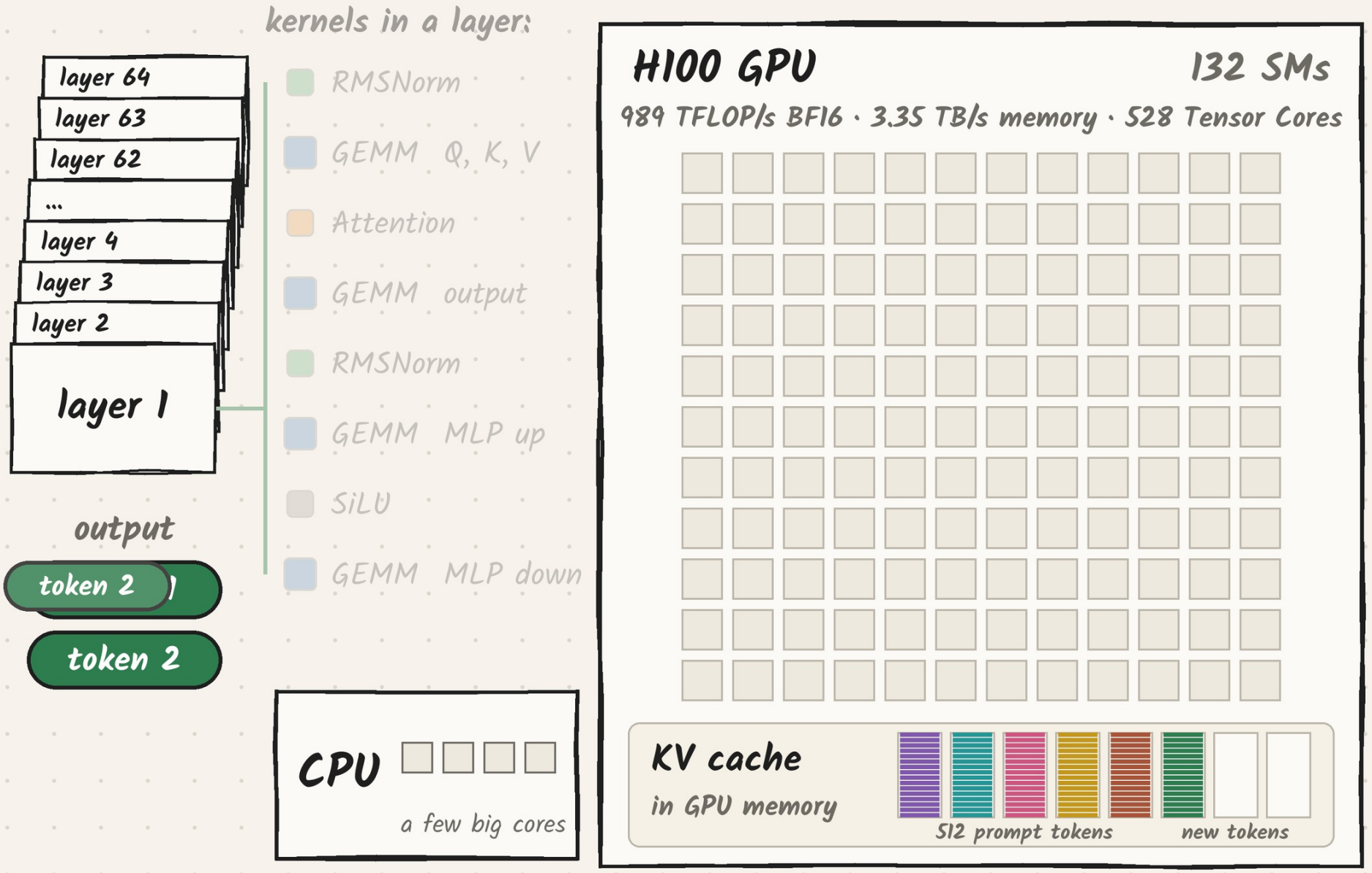
Output head picks token 1

prompt phase: 512 tokens in 54.3 ms

Qwen3.8-27B · vLLM · 1x H100 SXM · specs: NVIDIA H100 datasheet, Qwen config & tokenizer

AI through the lens of a GPU

how Qwen3.8-27B makes tokens on an H100



12/13

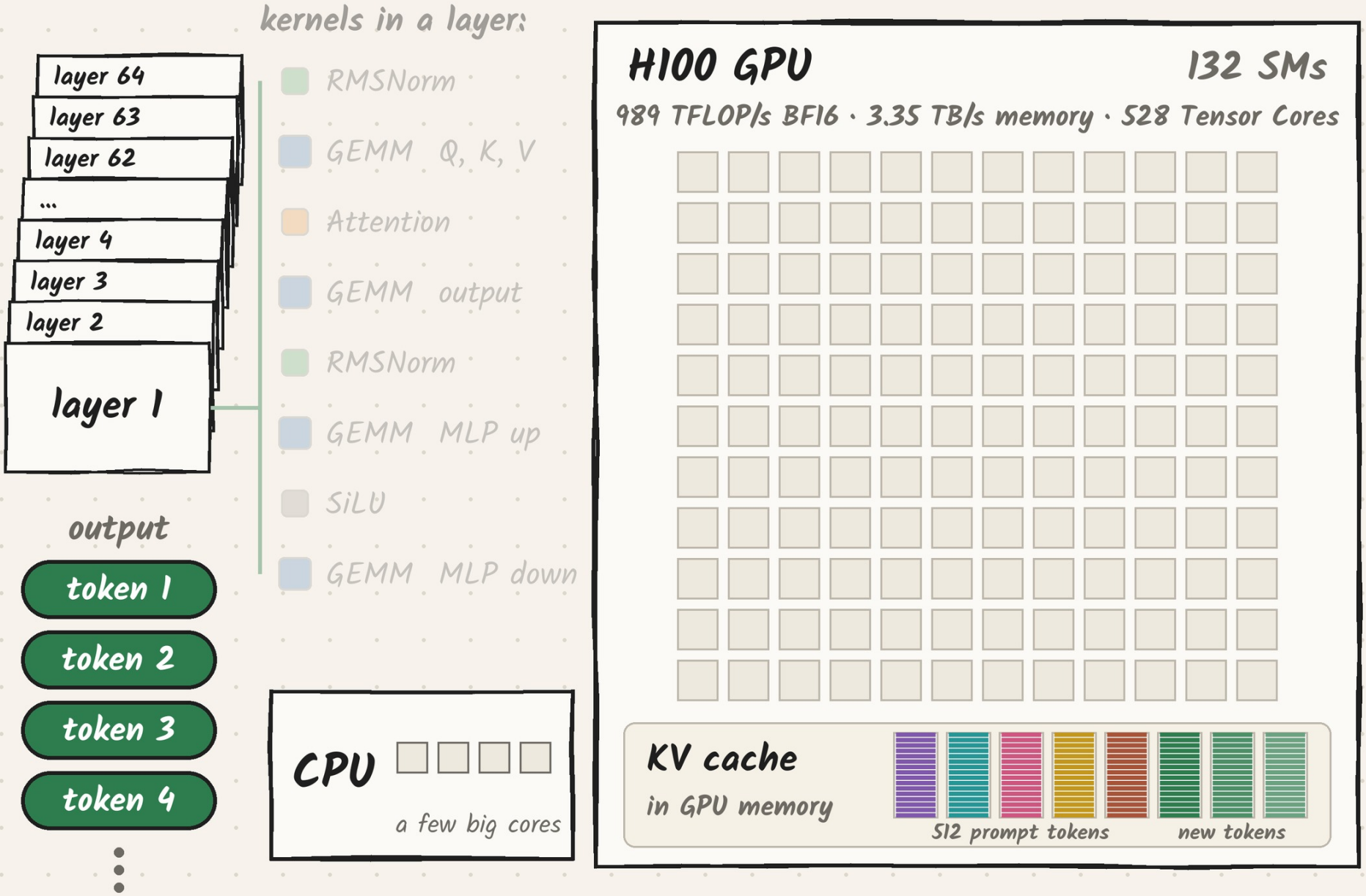
Decode: only the new token goes in

K, V of the 512 prompt tokens: KV cache

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13/13

One decode step: 902 kernels

20.55 ms per token, 48.7 tokens/s

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